AI PRODUCT INTELLIGENCE & FEEDBACK ANALYZER

Empirical Evaluation: Sparse N-Gram (TF-IDF) vs. Dense Semantic (MiniLM) Clustering

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thousands of tickets, reviews, and bug reports weekly.

1. PROBLEM & RESEARCH QUESTION

- Research Question: Does semantic embedding-based clustering group differently worded product feedback more usefully than a traditional TF-IDF-based approach?
- Vocabulary Mismatch: Customers use colloquial terms ('freezes on export' vs 'application shutdown during download').

Hypothesis: Dense continuous geometry prevents ticket fragmentation without requiring expensive commercial cloud LLM APIs.

2. SYSTEM ARCHITECTURE & WORKFLOW

A 4-tier modular pipeline with strict evaluation label isolation:

1. Ingestion: Multi-channel input, deduplication, URL context extraction.
2. Parallel Vectorization: TF-IDF (1-2 n-grams) vs. MiniLM (384-dim).
3. K-Means++ Clustering: Partitioning across k in [3, 4, 5, 6].
4. Decision Support: Negative Impact Score, rules & quote grounding.

Ingest & Clean	Dual Vectors	K-Means++	Action Cards

3. RIGOROUS EXPERIMENTAL SETUP

- Controlled Benchmark: Identical feedback records (N=20) evaluated across 12 configurations (k in [3..6], seeds: 42, 100, 2024).
- Strict Label Isolation: 'evaluation_labels.csv' kept physically separate; zero label leakage into vectorizers or clustering.
- Multi-Metric Harness:
  - External Agreement: Adjusted Rand Index (ARI) & NMI vs. 10 themes.
  - Pairwise Grouping: Precision, Recall, F1 over 190 record pairs.
  - Computational Latency: Sub-millisecond timing on CPU.
- Clean Install Verification: 10 automated health checks passed.

4. EVIDENCE-GROUNDED DECISION SUPPORT

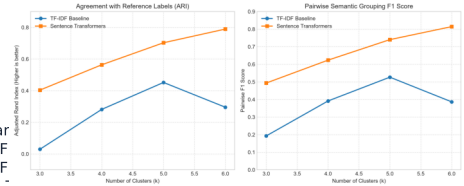
- Negative Impact Score = Size * (5 - Avg Rating) * Bug Ratio. Prioritizes widespread technical regressions over cosmetic feedback.
- Traceable Explainability: Centroid distance distributions and verbatim representative quotes anchor every recommended action.
- Persistence: Fully local SQLite database for history & auditing.

5. CONTROLLED EXPERIMENTAL RESULTS

Averaged across 12 runs (Seeds: 42, 100, 2024 | k in [3, 4, 5,

Model	Mean ARI	Pairwise F1	Latency
TF-IDF + K-Means	0.2647	0.3744	64.4 ms
Sentence Transformers	**0.6149**	**0.6680**	**62.8 ms***

*Note: MiniLM warmed inference is 62.8 ms (initial cold-start: 19.6s).
Key Finding: Sentence Transformers achieved +132% higher ARI and perfect 1.0 Pairwise Recall (19/19 pairs), uniting differently worded items.



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  - TF-IDF Result: Split into Cluster 3 & Cluster 2 (catch-all).
  - MiniLM Result: Correctly united into Cluster 0 (purity 1.0).

6. QUALITATIVE ERROR ANALYSIS

The TF-IDF Catch-All Problem:
- At k=4, TF-IDF Cluster 2 absorbed 11/20 records (55%), mixing auth timeouts, 404 bugs, and dark mode due to vector orthogonality.

7. THREATS TO VALIDITY & CONCLUSION

- Threats to Validity:
  - Synthetic dataset (N=20) designed for paraphrase evaluation; empirical replication on large customer telemetry is needed.
  - MiniLM groups by topic, not sentiment (speed praise + bug).
- Conclusion:
  - Dense embeddings definitively resolve vocabulary mismatch.
  - TF-IDF remains an essential baseline for sub-100ms triage and isolating rare keyword outliers (billing issues).
  - Optimal enterprise architecture is hybrid.

Refs: Reimers & Gurevych (EMNLP 2019); Salton & Buckley (1988); MacQueen (1967); Hubert & Arabie (1985). Zero paid cloud APIs.